

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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Frequency domain analysis of discrete non periodic signals

- Frequency domain representation of discrete non-periodic signals
- Spectrum of discrete non-periodic Signals
- Properties of discrete time Fourier transform



Non-periodic sequences that meet certain convergence conditions $x[k]$

$$x[k]=\frac{1}{2\pi}\int_{<2\pi>}X(\mathrm{e}^{\mathrm{j}\Omega})\cdot\mathrm{e}^{\mathrm{j}\Omega k}\,\mathrm{d}\Omega$$

The weighting function $X(\mathrm{e}^{\mathrm{j}\Omega})$ is called the frequency spectrum of a discrete non-periodic signal $x[k]$.

$$X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=-\infty}^{\infty}x[k]\mathrm{e}^{-\mathrm{j}\Omega k}$$

Frequency domain analysis of discrete non periodic signals

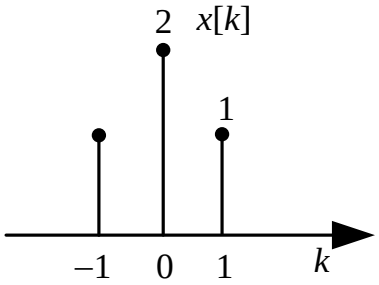
IDTFT $$x[k] = \frac{1}{2\pi} \int_{<2\pi>} X(e^{j\Omega}) \cdot e^{j\Omega k} \, d\Omega$$

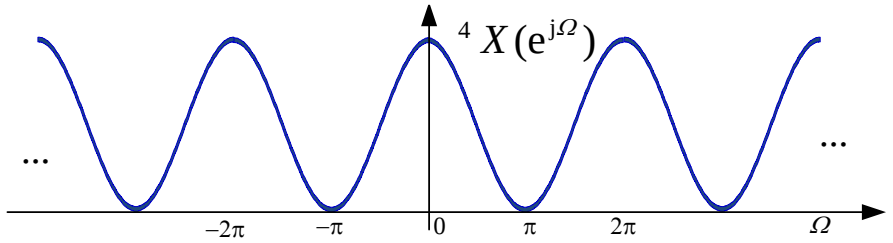
DTFT $$X(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} x[k] e^{-j\Omega k}$$

$$x[k] \xleftarrow{\text{DTFT}} X(e^{j\Omega})$$

Frequency domain analysis of discrete non periodic signals

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$X(e^{j\Omega})$ characteristic:

(1) $X(e^{j\Omega})$ is a continuous function of Ω

(2) $X(e^{j\Omega})$ is a periodic function with a period of 2π

$$X[e^{j(\Omega+2\pi)}]=\sum_{k=-\infty}^{\infty}x[k]e^{-j(\Omega+2\pi)k}=\sum_{k=-\infty}^{\infty}x[k]e^{-j\Omega k}\cdot e^{-j2\pi k}=X(e^{j\Omega})$$

Representation of Sequence Spectrum

$$X(e^{j\Omega}) = |X(e^{j\Omega})| e^{j\varphi(\Omega)}$$

amplitude spectrum

phase spectrum

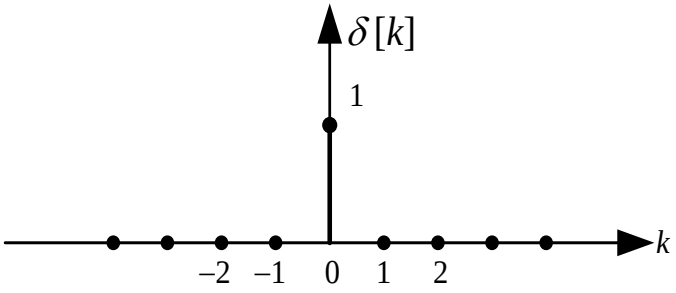
The principal value interval of the phase spectrum $\varphi(\Omega)$ is

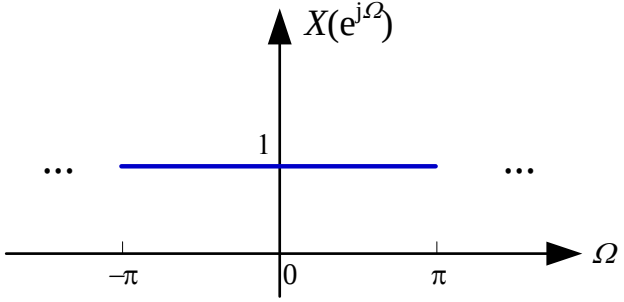
$$-\pi < \varphi(\Omega) \leq \pi$$

[E.g] Attempt to obtain the frequency spectrum of a unit pulse sequence $\delta[k]$

Solution: Defined by the DTFT of the sequence:

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-\infty}^{\infty} \delta[k]\mathrm{e}^{-\mathrm{j}\Omega k} = 1$$



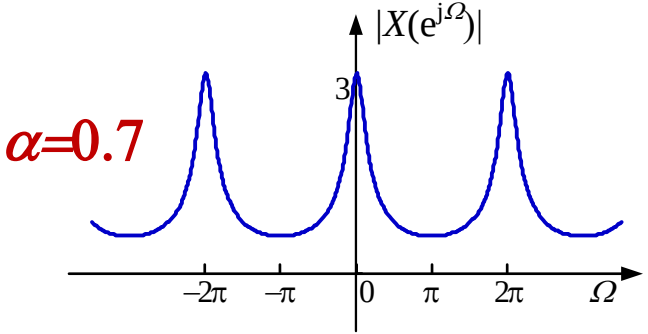


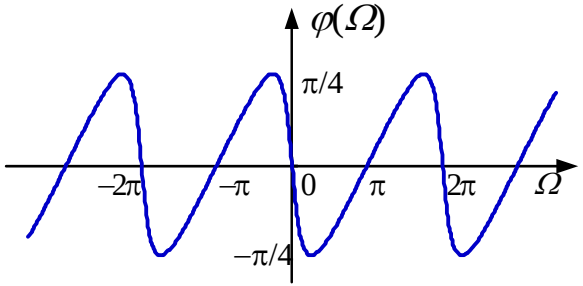
[E.g] Try to find the frequency spectrum of the sequence $x[k]=\alpha^k u[k]$

Solution: $X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=0}^{\infty}\alpha^k\mathrm{e}^{-\mathrm{j}\Omega k}=\sum_{k=0}^{\infty}\left(\alpha\mathrm{e}^{-\mathrm{j}\Omega}\right)^k$

When $|\alpha|>1$, Not satisfying absolute summation, the DTFT of the sequence does not exist.

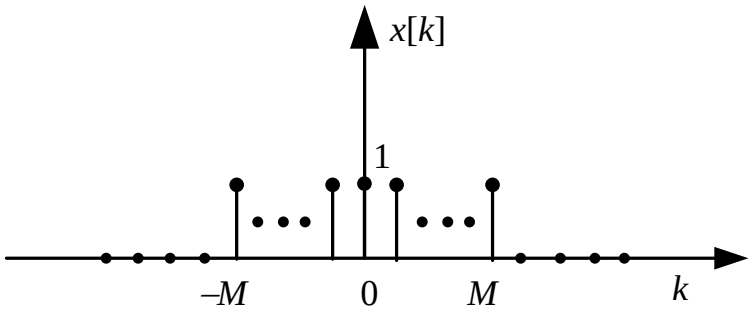
When $|\alpha|<1$, $X(\mathrm{e}^{\mathrm{j}\Omega})=\dfrac{1}{1-\alpha\,\mathrm{e}^{-\mathrm{j}\Omega}}$





[E.g] Find the frequency spectrum of a rectangular sequence $x[k]$ with a width of $2M+1$

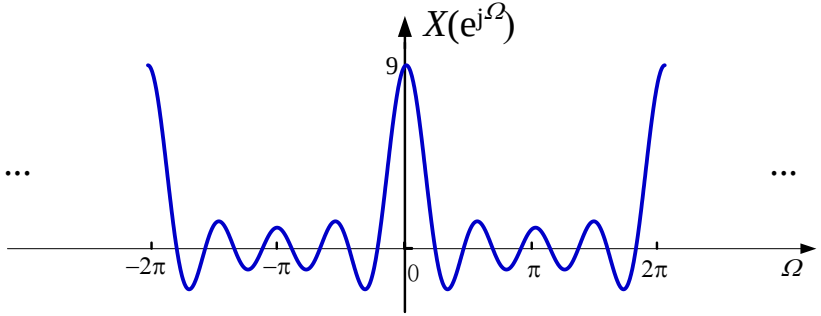
Solution: $$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k} = \sum_{k=-M}^{M} \left(\mathrm{e}^{-\mathrm{j}\Omega}\right)^k = \frac{\sin\left[\left(M+1/2\right)\Omega\right]}{\sin(\Omega/2)}$$



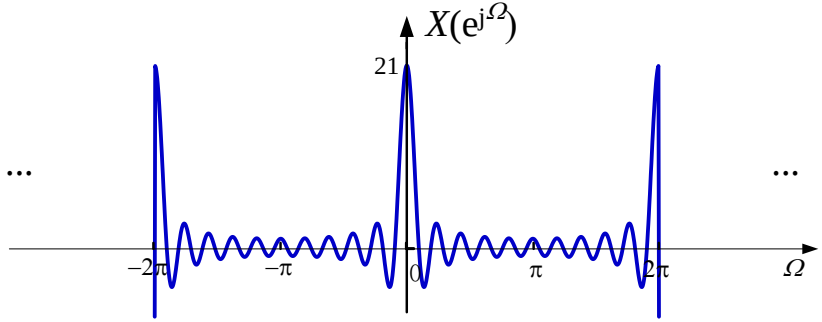
Rectangular sequence $x[k]$

[E.g] Find the frequency spectrum of a rectangular sequence $x[k]$ with a width of $2M+1$

Solution: $$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k} = \sum_{k=-M}^{M} \left(\mathrm{e}^{-\mathrm{j}\Omega}\right)^k = \frac{\sin\left[\left(M+1/2\right)\Omega\right]}{\sin(\Omega/2)}$$



Spectrum of rectangular sequence with M=4

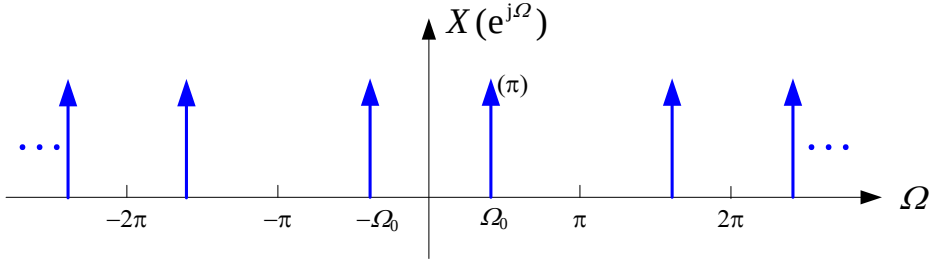


Spectrum of rectangular sequence with M=10

Given that the spectrum $X(e^{j\Omega})$ of a certain sequence $x[k]$ is as follows, try to find：

$$X(e^{j\Omega})=\pi\sum_{n=-\infty}^{\infty}[\delta(\Omega-\Omega_0-2\pi n)+\delta(\Omega+\Omega_0-2\pi n)]$$

Solution： According to the definition of IDTFT:



$$x[k]=\frac{1}{2\pi}\int_{<2\pi>}X(e^{j\Omega})e^{j\Omega k}d\Omega$$

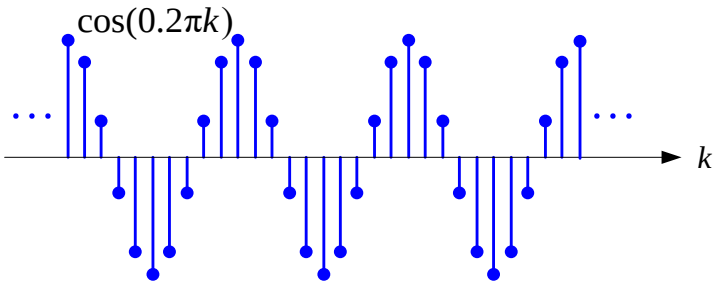
$$=\frac{1}{2}\int_{-\pi}^{\pi}[\delta(\Omega-\Omega_0)+\delta(\Omega+\Omega_0)]e^{j\Omega k}d\Omega$$

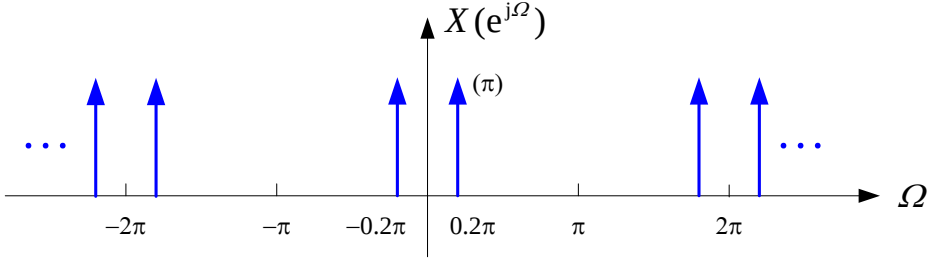
Given that the spectrum $X(e^{j\Omega})$ of a certain sequence $x[k]$ is as follows, try to find:

$$X(e^{j\Omega})=\pi\sum_{n=-\infty}^{\infty}[\delta(\Omega-\Omega_0-2\pi n)+\delta(\Omega+\Omega_0-2\pi n)]$$

Solutioin: $$x[k]=\frac{1}{2}\int_{-\pi}^{\pi}[\delta(\Omega-\Omega_0)+\delta(\Omega+\Omega_0)]e^{j\Omega k}d\Omega$$

$$=\frac{1}{2}(e^{j\Omega_0 k}+e^{-j\Omega_0 k})=\cos(\Omega_0 k)$$





Properties of Discrete Time Fourier Transform

※ Linear characteristics

※ Symmetrical characteristics

※ Shift characteristics

※ Convolutional properties

※ Parseval theorem

※ Linear characteristics

If $x_1[k] \xleftarrow{DTFT} X_1(e^{j\Omega})$

Then $x_2[k] \xleftarrow{DTFT} X_2(e^{j\Omega})$

$$a x_1[k] + b x_2[k] \xleftarrow{DTFT} a X_1(e^{j\Omega}) + b X_2(e^{j\Omega})$$

Symmetrical characteristics

$$x^*[k] \xleftarrow{\text{DTFT}} X^*(e^{-j\Omega}) \qquad x^*[-k] \xleftarrow{\text{DTFT}} X^*(e^{j\Omega})$$

When $x[k]$ is a real sequence

$$X(e^{j\Omega}) = X^*(e^{-j\Omega})$$

Amplitude and phase

$$|X(e^{j\Omega})| = |X(e^{-j\Omega})|$$

$$\varphi(\Omega) = -\varphi(-\Omega)$$

Real and imaginary parts

$$X_R(e^{j\Omega}) = X_R(e^{-j\Omega})$$

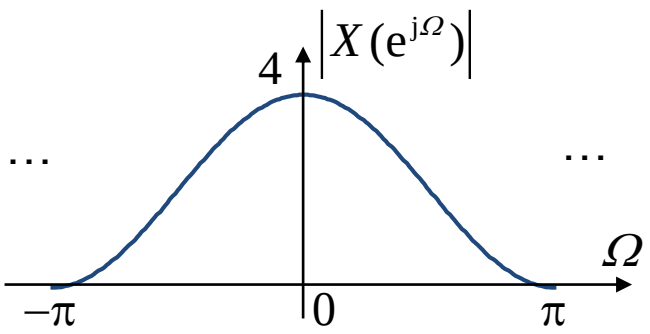
$$X_I(e^{j\Omega}) = -X_I(e^{-j\Omega})$$

[E.g] Find the amplitude spectrum and phase spectrum of sequence $x[k]=\{1,2,1;\ k=0,1,2\}$.

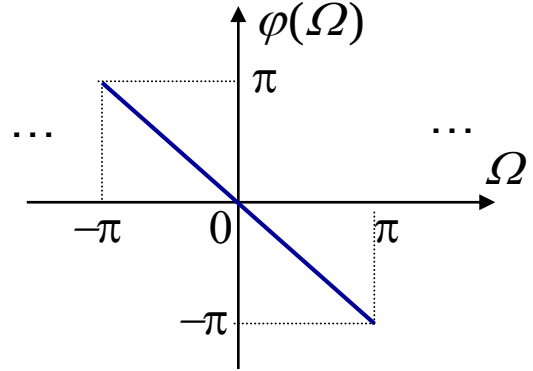
Solution: $$X(\mathrm{e}^{\mathrm{j}\Omega})=1+2\mathrm{e}^{-\mathrm{j}\Omega}+\mathrm{e}^{-\mathrm{j}2\Omega}=(1+\mathrm{e}^{-\mathrm{j}\Omega})^2=[\mathrm{e}^{-\mathrm{j}\frac{\Omega}{2}}(\mathrm{e}^{\mathrm{j}\frac{\Omega}{2}}+\mathrm{e}^{-\mathrm{j}\frac{\Omega}{2}})]^2$$

$$=[\mathrm{e}^{-\mathrm{j}\frac{\Omega}{2}}2\cos\frac{\Omega}{2}]^2=\mathrm{e}^{-\mathrm{j}\Omega}4\cos^2\frac{\Omega}{2}$$

amplitude spectrum: $\left|X(\mathrm{e}^{\mathrm{j}\Omega})\right|=4\cos^2\frac{\Omega}{2}$ phase spectrum: $\varphi(\Omega)=-\Omega$



Amplitude spectral symmetry



Phase spectrum odd symmetry

[E.g] Find the DTFT of the sequence $y[k]=\alpha^{|k|},|\alpha|<1$

Solution: Utilize symmetry and linear properties

Due to $$x[k]=\alpha^{k}u[k]\xleftrightarrow{\text{DTFT}}X(\mathrm{e}^{\mathrm{j}\Omega})=\frac{1}{1-\alpha\mathrm{e}^{-\mathrm{j}\Omega}}$$

$$\because\quad y[k]=\alpha^{k}u[k]+\alpha^{-k}u[-k]-\delta[k]=x[k]+x^{*}[-k]-\delta[k]$$

$$\therefore\quad \mathrm{DTFT}\{y[k]\}=X(\mathrm{e}^{\mathrm{j}\Omega})+X^{*}(\mathrm{e}^{\mathrm{j}\Omega})-1=2\,\mathrm{Re}\{X(\mathrm{e}^{\mathrm{j}\Omega})\}-1$$

$$=\frac{1-\alpha^{2}}{1-2\alpha\cos\Omega+\alpha^{2}}$$

Properties of Discrete Time Fourier Transform

◆ Shift characteristics

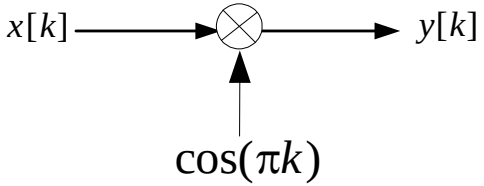
Time shift characteristics: $x[k-n] \xleftarrow{DTFT} e^{-j\Omega n} X(e^{j\Omega})$

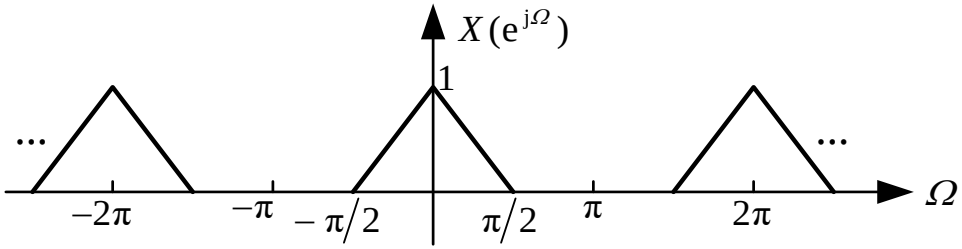
Frequency shift characteristics: $e^{j\Omega_0 k} x[k] \xleftarrow{DTFT} X(e^{j(\Omega-\Omega_0)})$

The time-domain time shift of a sequence corresponds to the frequency-domain phase shift.

The time-domain phase shift of a sequence corresponds to the frequency-domain frequency shift.

[E.g] Given the spectrum of $x[k]$ as shown in the figure, try to find the spectrum of $y[k]=x[k]\cos(\pi k)$



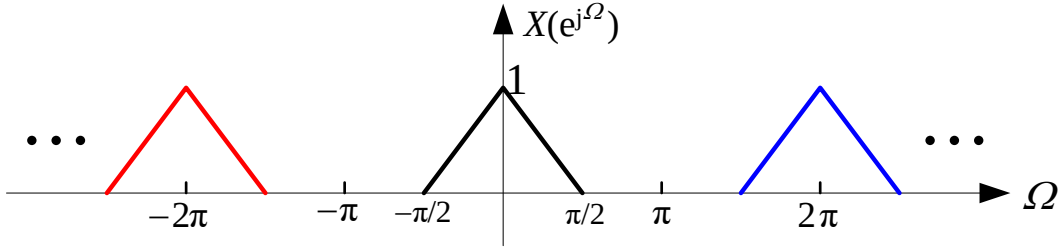


Solution: Utilize frequency shift characteristics

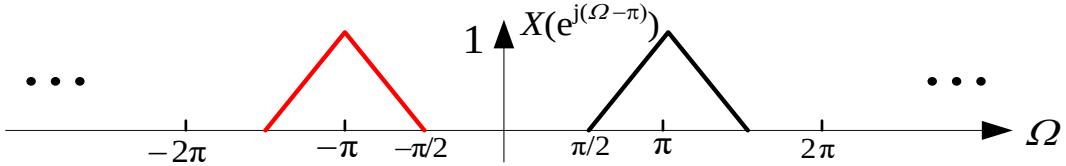
$$y[k]=x[k]\cos(\pi k)=x[k]\left(\frac{e^{j\pi k}+e^{-j\pi k}}{2}\right)$$

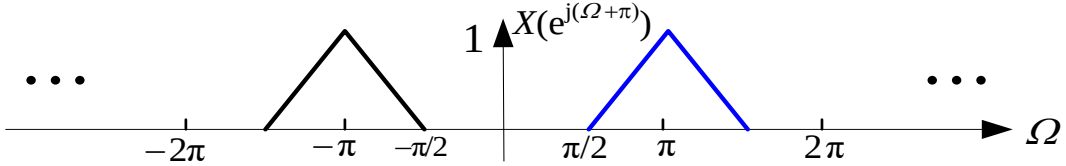
$$Y(e^{j\Omega})=\frac{X(e^{j(\Omega-\pi)})+X(e^{j(\Omega+\pi)})}{2}$$

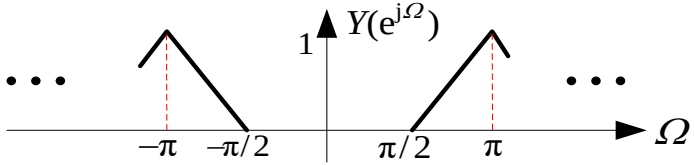
[E.g] Given the spectrum of $x[k]$ as shown in the figure, try to find the spectrum of $y[k]=x[k]\cos(\pi k)$



Solution: $Y(e^{j\Omega}) = \dfrac{X(e^{j(\Omega-\pi)}) + X(e^{j(\Omega+\pi)})}{2}$







Properties of Discrete Time Fourier Transform

The convolution property

$$x[k] * h[k] \xleftarrow{\text{DTFT}} \longrightarrow X(e^{j\Omega})H(e^{j\Omega})$$

$$x[k]h[k] \xleftarrow{\text{DTFT}} \longrightarrow \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})H(e^{j(\Omega-\theta)})d\theta$$

The convolution of sequence in time domain corresponds to the product of frequency domain.

The product of sequence time domain corresponds to periodic convolution in frequency domain.

Properties of Discrete Time Fourier Transform

◆ Parseval theorem

$$\sum_{k=-\infty}^{\infty} |x[k]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| X(e^{j\Omega}) \right|^2 d\Omega$$

The energy in the time domain of a sequence is equal to the energy in the frequency domain

$$\sum_{k=-\infty}^{\infty} x[k]x^*[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) X^*(e^{j\Omega}) d\Omega$$

E.g: Given a finite length sequence $x[k]=\{1,4,3,2;\ k=0,1,2,3\}$, do not calculate the spectrum $X(e^{j\Omega})$ of $x[k]$ directly determines the value of the following expression.

Solution:

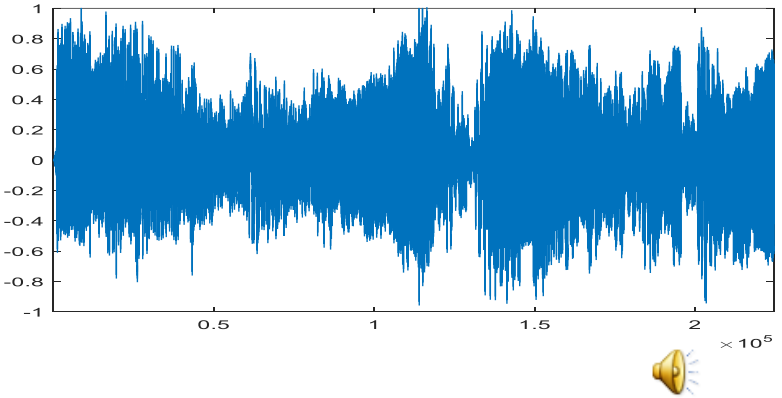
(1) $$X(e^{j0})=\sum_{k=0}^{3}x[k]\cdot e^{-jk0}=\sum_{k=0}^{3}x[k]=1+4+3+2=10$$

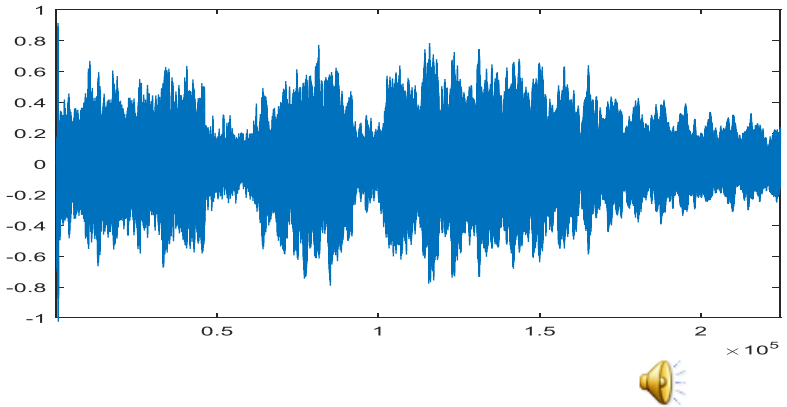
(2) $$X(e^{j\pi})=\sum_{k=0}^{3}x[k]\cdot e^{-jk\pi}=\sum_{k=0}^{3}x[k]\cdot(-1)^{k}=1-4+3-2=-2$$

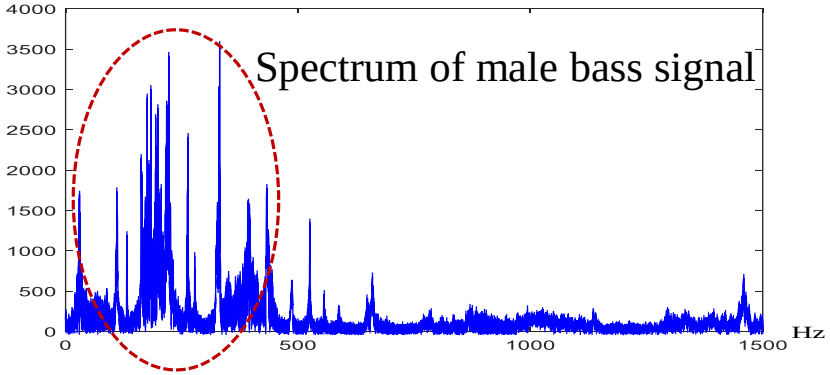
(3) $$\int_{-\pi}^{\pi}X(e^{j\Omega})\,d\Omega=2\pi x[0]=2\pi$$

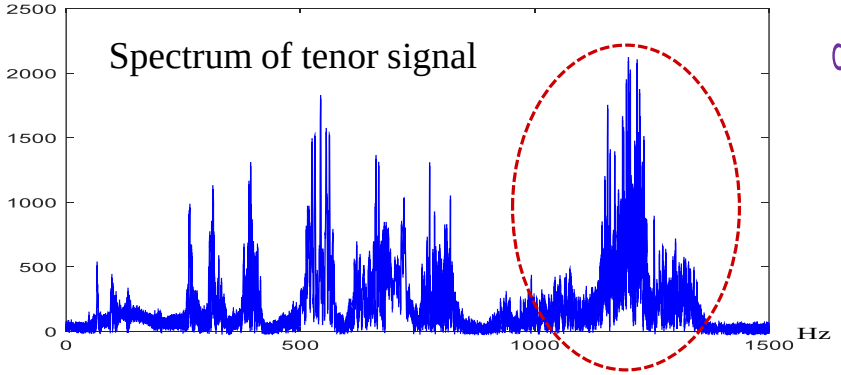
(4) $$\int_{-\pi}^{\pi}\left|X(e^{j\Omega})\right|^{2}d\Omega=2\pi\sum_{k=0}^{3}\left|x[k]\right|^{2}=60\pi$$

Application examples of signal frequency domain analysis









Analyze sounds in different registers

E.g Given a finite length sequence $x[k]=\{1,3,3,1;\ k=0,1,2,3\}$,

calculate the amplitude spectrum of $x[k]$.

Solution: $X(\mathrm{e}^{\mathrm{j}\Omega})=\sum_{k=0}^{3}x[k]\cdot\mathrm{e}^{-\mathrm{j}k\Omega}$

$$=1+3\mathrm{e}^{-\mathrm{j}\Omega}+3\mathrm{e}^{-\mathrm{j}2\Omega}+\mathrm{e}^{-\mathrm{j}3\Omega}$$

$$=\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}(\mathrm{e}^{\mathrm{j}\frac{3}{2}\Omega}+\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega})+3\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}(\mathrm{e}^{\mathrm{j}\frac{1}{2}\Omega}+\mathrm{e}^{-\mathrm{j}\frac{1}{2}\Omega})$$

$$=2\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}\cos(\frac{3}{2}\Omega)+6\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}\cos(\frac{1}{2}\Omega)$$

$$\left|X(\mathrm{e}^{\mathrm{j}\Omega})\right|=\left|2\cos(\frac{3}{2}\Omega)+6\cos(\frac{1}{2}\Omega)\right|$$

E.g Calculate the discrete sequence $x[k]$ corresponding to the following spectral functions.

$$X_1(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{n=-\infty}^{\infty} \delta(\Omega + 2\pi n) \qquad X_2(\mathrm{e}^{\mathrm{j}\Omega}) = 1 + 2\sum_{n=1}^{N} \cos(\Omega n)$$

Solution: $x_1[k] = \dfrac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\mathrm{e}^{\mathrm{j}\Omega})\mathrm{e}^{\mathrm{j}k\Omega}\mathrm{d}\Omega = \dfrac{1}{2\pi}\int_{-\pi}^{\pi} \delta(\Omega)\mathrm{e}^{\mathrm{j}k\Omega}\mathrm{d}\Omega = \dfrac{1}{2\pi}$

$$X_2(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{e}^{\mathrm{j}N\Omega} + \mathrm{e}^{\mathrm{j}(N-1)\Omega} + \cdots + \mathrm{e}^{\mathrm{j}\Omega} + 1 + \mathrm{e}^{-\mathrm{j}\Omega} + \cdots + \mathrm{e}^{-\mathrm{j}(N-1)\Omega} + \mathrm{e}^{-\mathrm{j}N\Omega}$$

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \sum_{k=-\infty}^{\infty} x[k]\mathrm{e}^{-\mathrm{j}\Omega k} \qquad x_2[k] = \begin{cases} 1 & |k| \le N \\ 0 & \text{其他} \end{cases}$$

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.